

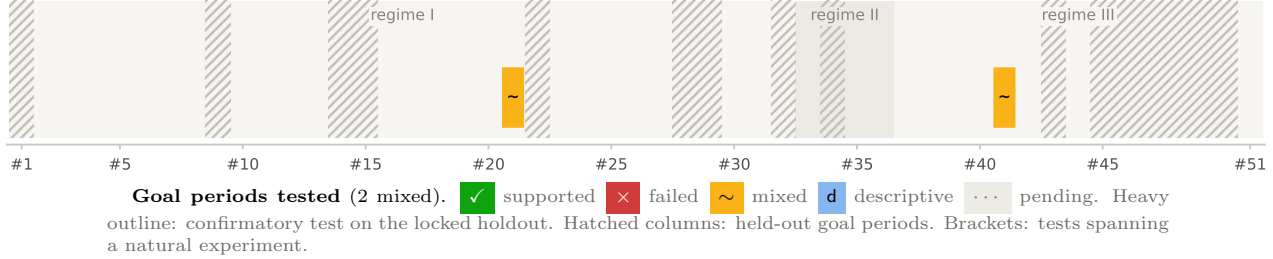
H24: In the forecast week (#21), switching from independent drafts to comparison switches the coupling on

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 1.5/5

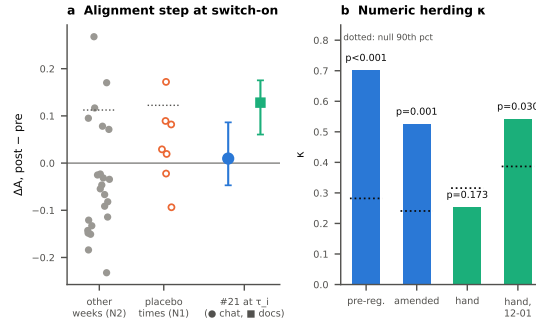
Bottom line. The forecast-week coupling switch moved swarm alignment no more than ordinary kickoff days; coupling is pairwise, about 0.03 in cosine per document read.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Goal period #21 (AI forecasts; regime I; 8–9 agents) serves as a natural experiment. The record dates the first reading of a teammate’s forecast at 42–86 min into the week (median 50 min), not mid-week. There, the share of messages that name a teammate jumps from 7% to 74%. The predicted alignment step is absent: $\Delta A_{\text{res}} = +0.010 [-0.05, +0.09]$, inside a within-week placebo and 23 other kickoff days. Content was aligned before any reading ($A_{\text{res}} \approx 0.33$ vs a rotation null of 0.03). Numeric herding was an extraction artifact (hand-verified $\kappa = 0.25$, $p = 0.17$). **Round 1b** adds a second embedding model, shared and multi-direction goal fields, dedupe and style residuals. The step fails in all 13 configurations. gte sees +0.16, but ordinary kickoff days match it. Natives: reading a teammate’s document pulls the reader toward it by ≈ 0.03 . One kickoff split into two rooms grew two room cultures. The holdout (#14) is not run.



Observables. (a) Change in residual alignment across the switch-on in #21 (blue: statements; green: documents; 90% CIs). Comparators: 23 other regime-I kickoff days at the same offset (gray) and placebo switch times inside #21 (orange). Dotted lines: 90th percentiles. (b) DeGroot pull κ under the pre-registered extractor, the amended extractor and hand-verified values. Dotted: the 90th percentile of the independent-updating null. The herding signal shrinks to noise once extraction errors are removed.

PHYSICAL MODEL

Vector spins (11) with a Curie–Weiss analogue (01). Spin $\mathbf{s}_i$: the whitened, unit-normalized statement embedding of agent i ($n = 32$). Field $h\hat{g}$: the goal and kickoff text. $\mathbf{h}_i$: agent style. Coupling J_0 : a mean-field pull toward the others, switched on at τ_i , the first reading of a teammate’s forecast by agent i . Order parameter A : mean pairwise cosine; A_{res} removes $\hat{g}$. Numeric forecasts x_i (AGI by 2035, SI by 2050, p(doom)) are soft spins with a DeGroot pull κ .

$$\mathbf{s}_i(t) \propto h\hat{g} + \mathbf{h}_i + \boldsymbol{\xi}_i(t) + J_0 \Theta(t - \tau_i) \mathbf{m}_{-i}(t)$$

$$A = \frac{2}{N(N-1)} \sum_{i < j} \hat{\mathbf{v}}_i \cdot \hat{\mathbf{v}}_j, \quad \beta J_0/n = 1 - \frac{1}{1 + (N-1)A_{\text{res}}}$$

$$\Delta x_i = \kappa (\bar{x}_{-i} - x_i) + \epsilon_i, \quad \kappa = 0 \text{ for independent updating}$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An “independent drafts first” phase does not protect the diversity of swarm forecasts. Content was aligned before any agent read another ($A_{\text{res}} \approx 0.33$), from the shared prompt, instructions and prior. Measure diversity at kickoff, not after drafting. Comparison then produced talk about divergence and some literal copying of numbers, not an alignment jump. Do not measure herding on regex-extracted numbers: extraction errors make false herding through regression to the mean. One week with $N = 7$ and no ground truth makes this a pilot.

RESULTS VS PREDICTIONS

Prediction	Round 1 → 1b (bge / gte)	Verdict
P1 step > q90 of N1 and N2	+0.010 → -0.011 / +0.160 (N2 q90 × 0.10 / 0.18); 13/13 fail	×
P2 documents $\Delta A_{\text{res}} > 0$	+0.13 → +0.16 / +0.08 (CIs > 0)	✓
P3 $\beta J_0/n$ rises	0.67→0.67 → flat / 0.69→0.76	model-dep.
P4 numeric herding (audited)	κ 0.25, p 0.17 (no embedding)	×
P6 ramp $\rho > 0$	0.73 → 0.47 / 0.60	~
Levels vs rotation null	0.33 vs 0.03 → 0.34–0.42 vs ≤0.10	✓
#21 reading DiD (na-tive)	statements +0.036 (p 0.035) / +0.030 (p 0.038); style p 0.08–0.14	✓
#41 two rooms (na-tive)	no gap in first drafts; gap grows to +0.25 / +0.32 by day 4	~

GOAL PERIODS IN DETAIL

G21 ~ 1b: mixed (step fails; pair DiD supported)
G41 ~ mixed (round 1b native; both models)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	0	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

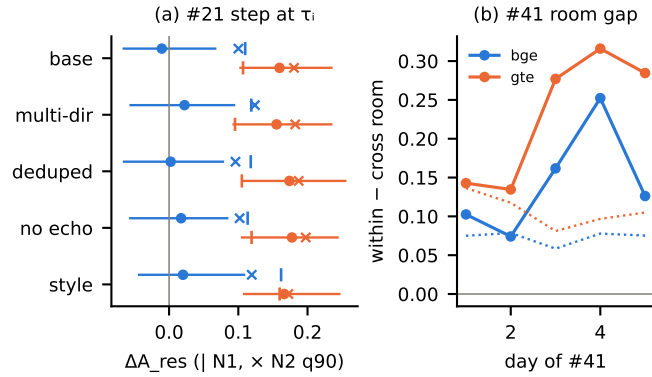
CAVEATS

The test bed is one forecasting week with $N = 7$. The switch-on and the reading events rest on the agents' own screen descriptions. The step statistic depends on the embedding model. The pair DiD weakens after style residualization, so part of it may be register. In #41 everyone reads roommates within a minute, so early coupling and room composition are not separable. #41 has only 9 one-statement drafts.

ROUND 2

Measure the copy fidelity of numbers after reading (H07 machinery). Test whether first-posted numbers dominate the consensus (anchoring), per the round-2 redirects.

- **Essence.** Information aggregation in swarms is copying from the first or most confident agent, not averaging.
- **H24-R1** Anchoring cascade: first-posted numbers dominate the final consensus (a founder effect) in every estimation-like task.
- **H24-R2** Copy fidelity vs transformation of shared numbers (H07 machinery).



Second figure. (a) Step at the switch-on per input: blue bge, orange gte, bars 90% CI. Markers: | within-week placebo q90, × kickoff-matched q90. gte steps up, but no more than ordinary kickoff days. (b) #41: within-room minus cross-room alignment by day (dotted: permutation q95).